SIXTH SEMESTER [B.TECH]

PRINCIPLES OF MANAGEMENT FOR ENGINEERS [MS-302]

UNIT - I

Q.1. What is Management? Discuss in detail evolution of management. What is relevance of contingency theory of management?

Ans. Management is the process of planning and organizing the resources and activities of a business to achieve specific goals in the most effective and efficient manner possible. Efficiency in management refers to the completion of tasks correctly and at minimal costs. Effectiveness in management relates to the completion of tasks within specific timelines to yield tangible results.

Characteristics of management

Some of the fundamental characteristics of management are as follows:

Multi-dimensional

Most management oversees and supervises a company or organization's service or production cycle. Managers work closely with and provide guidance to the members of their team. A manager considers a staff member both as an individual with diverse needs and as a component of the larger group. To be effective, managers influence their team members to apply their unique strengths toward achieving the organization's goals.

Dynamic

Management is a dynamic function and evolves and adapts to changes in its environment, whether they are economic, socio-political or technological. For instance, a paper company could see a decline in sales because of the rapid adoption of screens and digital devices. Whether the company can still survive depends on how effectively its management can adapt to new market requirements.

Intangible

Management is not a tangible product, but its presence can change the way an organization functions. Management consists of ideologies, policies and human interaction. Good management helps improve a company's target achievement ratios, employee gratification levels and overall ease in the company's operation.

Evolution of Management

Classical Management theories

Classical management theory prioritizes profit and assumes that personal gain motivates employees. It aims to streamline operations and increase productivity.

Major concepts include specialization, incentivization, and hierarchical structure. The first two contribute to employee efficiency and drive. Centralized leadership simplifies decision-making, and a meritocratic chain of commands provides order and oversight. At every level, standardization reduces waste and error.

Scientific Management Theory

Scientific management theory is sometimes called Taylorism after its founder Frederick Winslow Taylor, a mechanical engineer. Taylor employed scientific methods to develop organizational principles that suited mass production needs. After creating and proving his theory as a manager and consultant, he wrote " The Principles of Scientific Management" in 1911.

Bureaucratic Management Theory

Max Weber was one of the foremost scholars of the late 19th and early 20th century. He strongly influenced – and continues to influence – economic, religious, and political sociology. He explains bureaucratic management theory in "Economy and Society," published posthumously in 1922.

Weber believed that standard rules and well-defined roles maximize the efficiency of an organization. Everyone should understand the responsibilities and expectations of their position, their place within a clear hierarchy and general corporate policies. Hiring decisions and the application of rules should be impersonal, guided only by reason and established codes.

Administrative Management Theory

Just as scientific management theory is sometimes called Taylorism, administrative management theory is sometimes called Fayolism.

Henri Fayol was a mining engineer who sought to codify the responsibilities of management and the principles of effective administration. He outlined these in "General and Industrial Management" in 1916.

His guide identifies 14 principles of management:

- Division of work: Divide work into tasks and between employees.
- Authority: Balance responsibility with commensurate authority.
- Unity of command: Give each employee one direct manager.
- Unity of direction: Align goals between employees.
- Equity: Treat all employees equally.
- Order: Maintain order through an organized workforce.
- Discipline: Establish and follow rules and regulations.
- Initiative: Encourage employees to show initiative.
- Remuneration: Pay employees fairly for the work they do.
- Stability: Ensure that employees feel secure in their positions.
- Scalar chain: Establish a clear hierarchy of command.
- Subordination of individual interest: Prioritize group needs.
- Esprit de corps: Inspire group unity and pride.

Relevance of contingency theory

The contingency theory is simply an extension of the systems organizational theory. According to this theory, there is no particular managerial action or organizational design that is appropriate for all situations.

In fact, the design, as well as the managerial decision, depends on the situation. In other words, it is contingent on the situation and circumstances. Therefore, the Contingency Theory is also referred to as a situational theory.

- The systems theory focuses on the internal dynamics of an organization's structure and behavior. On the other hand, the contingency organizational theory focuses on the external determinants of the organization's behavior and structure.
- The systems theory lays down universal principles for application in all situations. On the other hand, the contingency organizational theory works on the prescription which says that 'it all depends'.
- In other words, the contingency organizational theory spells out the relationship between an organization and its external environment and endeavors to fill a critical lacuna of the systems theory.

• Further, it offers a more explicit understanding of the relationship between various variables of the environment. It is also action-oriented and directed towards the application of the system theory's concepts.

• Hence, it offers useful and practical insights to managers in turbulent working environments.

Q.2. How an organizational culture and climate connected? How can a manager create and sustain organizational culture.

Ans. Although many companies and hiring managers use the phrases company culture and organizational climate interchangeably, it is important to note the differences so you can understand the reasons behind company employee behavior and decide if they are a right fit for you.

Company culture characteristics

These characteristics are unique to company culture.

• **Culture is abstract:** Company cultures are embedded belief systems held by employees and leadership. They are passed down and reveal themselves through policy and unwritten expectations.

• **Difficult to change:** Because it is composed of abstract regulations that govern behavior, company culture is static and very difficult to change. A positive culture attracts new talent, but if company culture is unhealthy, there will be considerable employee turnover and transfers.

• **Governs employee behavior:** Much of human behavior is dictated by the established norms of the community. Culture will guide the interactions and affect employees' ability to build professional relationships. A company culture that values professionalism will reveal itself through employee conduct.

• **Embedded in beliefs and actions:** Because it is hard to identify the embedded beliefs of a company that affect expectations and actions, it is hard to measure the effect of culture.

Company climate characteristics

The following characteristics differentiate company climate:

• Climate is based on employee opinion. Climate is the attitude employees have toward the workplace environment. Because different employees have unique experiences, the adapted attitude changes from employee to employee. If you are deciding whether a company is right for you, consider all aspects and perspectives before committing.

• Can be an emotional response. If an employee has experienced something positive, such as receiving an award for quality performance, their perspective on company climate will be positive. Coaching has become a common strategy in company's to ensure leadership feedback fosters a positive work environment, influencing a positive employee opinion.

• Linked to environment. The floor plan of the physical work environment, the decorations or lack thereof, and the office furniture all communicate a type of environment. Flexible seating arrangements with bean bags and movable furniture show collaboration and comfort are important, but heavy office chairs, thick couches, and large tables indicate set structures.

• Affects productivity and performance. An organization's climate directly affects employee job satisfaction, task completion and productivity, and job performance. If employers improve workplace climate, they can succeed in improving employee performance.

Ways to create positive organizational culture

Set clear departmental goals

Outline the objectives of each team so employees have tangible results to work toward. Not only will this help guide individual performance, but it will encourage collaboration between team members. Make sure there is room for feedback to adjust quotas and KPIs when needed.

For example, if a team is continually reaching their objectives without breaking a sweat, you might want to modify their target goals to push production further.

Promote the organization's goals

In addition to setting departmental goals, make sure every employee is clear on the organization's long-term objectives. This will help individuals cultivate a sense of professional purpose. Having a source of motivation beyond quarterly quotas will demonstrate the value each role has toward achieving the company's mission.

Promote diversity and inclusivity

Create a positive, inclusive work culture by welcoming individuals from all backgrounds and celebrating their differences. Encourage employees to share their pronouns with the rest of the team to promote inclusive language and consider establishing a committee to contribute to diversity initiatives. Work with the HR department to make diversity a part of your recruitment strategy and ensure diversity and inclusion continue to be foundational elements as your organization grows.

Allow for lightheartedness

Work has its stressful moments and being able to make a difficult situation more lighthearted is an invaluable skill. Of course, the ultimate goal should be to resolve the problem, but a fresh perspective and positive outlook is more productive than the alternative. As Dale Carnegie, an American writer and lecturer, said, "People rarely succeed unless they are having fun in what they are doing." If you can afford to find the bright side and let your team know that you have their back, they'll return the favor by working even harder.

Prioritize respect

Every individual should feel valued and heard, regardless of their status within the company. Interns offer a much greater advantage than being delegates for busy work, and new employees bring a fresh perspective. You never know where the next big idea will come from, so let every employee have a seat at the table and feel empowered to share their thoughts.

Establish a strict zero tolerance policy

Just as important as creating a welcoming environment is ensuring employees know their rights and individualities are protected within the workplace. A crucial facet of a positive work culture is providing employees with the opportunity to speak openly about issues they are facing–in and outside of the office–and have access to the support and resources they need. Make sure HR representatives have flexibility within their schedules to be available for personal conversations when needed, and consider implementing an anonymous sexual harassment hotline as a secure and private way for employees to report incidents in the workplace.

Create an employee recognition program

Recognize and reward employees for achieving outstanding results. Doing so will encourage employees to continue performing at impressive levels, and make them feel valued within the company. It will also motivate their peers to up their game, fostering a work culture of friendly competition that leads to high performance.

Q.3. Discuss different roles and functions of Manager?

Ans. Roles and responsibilities of a manager

Managers need the cooperation and skills of the people who work for them. The manager "wears many hats" to develop, motivate and drive their team:

Leader: The manager sets the vision and motivation for their team. They develop the plan and drive their team to achieve it. A good manager leads and delegates by earning the respect of employees and motivating them to be their best.

Project manager: A manager turns company goals into actionable project plans. They use budgets, resources and systems efficiently.

Coach: Managers train their employees properly and help them grow within the company. They help employees develop skills and their careers.

Resource planner: Managers hire new employees, address performance issues, and when needed, discipline or terminate employees. This is an integral part of the job, and when they use the proper hiring strategies, a manager can create a successful team of employees.

Functions of a manager are the various roles played by the manager in an organization. A manager is accountable for all the happenings in the firm and is answerable to the management. The seven major roles played by the manager are:

- Planning
- Organizing
- Staffing
- Directing/leading
- Coordinating
- Reporting
- Budgeting
- Controlling

Q.4. (a) "Management is an art of getting things done". Do you agree? Give reasons?

Ans. This definition of management does not integrate the growing role of technology in business processes and therefore in the management process. A better definition to convey the present-day concept of management would probably be something more along the lines of management is an art of getting things done through leveraging technology to maximize the productivity of human resources.

Almost every industry has gone through some phase of digitalization, which means that managers must decide how much to invest in technology to improve production, delivery, marketing and a range of other business applications. Even in many service industries that to the end consumer seemingly remain unchanged, there have been and continue to be technological upgrades that enable better forecasting and billing, as well as a myriad of other functions to improve profitability and facilitate business.

When a leader inspires his or her subordinates, it prompts them to work hard. More work is accomplished as a result, and the organization prospers. That's how managers get things done through people.

Modern organizations offer flexible working hours so that employees can work on personal projects during their free time. Such companies follow the management principles of Follet. They know that if they give workers liberty, they will be more productive.

In some ways, this statement is true. Successful management entails the recruitment, training, and supervision of individuals chosen for tasks specifically tailored toward company goals, so in this respect, interpersonal interaction is very important as a means of maintaining efficiency and cohesion within a modern marketplace that is constantly evolving to suit increasingly specific consumer demands. However, given recent technological advances in the business sector, business has become more and more depersonalized, thereby reducing the need for highly developed interpersonal skills within the workplace. Before computers, tasks were organized and carried out person to person, with very few intermediaries to facilitate these demands. Now, in the age of Big Data, it has become more and more necessary to acquire skills related to the management of computer software (Microsoft Excel) rather than skills relating to the management of people. So, while it is true that successful management still entails working well with others, it has become more and more important to get things done through the successful integration and mastery of technological advances that surface constantly within a continuously evolving economy.

Q.4. (b) Explain the managerial level in an organization?

Ans. LEVELS OF MANAGEMENT

(a) Top Level of Management

It consists of board of directors, chief executive or managing director.

The top management is the ultimate source of authority and it manages goals and policies for an enterprise. It devotes more time on planning and coordinating functions.

The role of the top management can be summarized as follows -

- Top management lays down the objectives and broad policies of the enterprise.
- It issues necessary instructions for preparation of department budgets, procedures, schedules etc.
- It prepares strategic plans & policies for the enterprise.
- It appoints the executive for middle level i.e. departmental managers.
- It controls & coordinates the activities of all the departments.

(b) Middle Level of Management

The branch managers and departmental managers constitute middle level. They are responsible to the top management for the functioning of their department. They devote more time to organizational and directional functions.

In small organization, there is only one layer of middle level of management but in big enterprises, there may be senior and junior middle level management. Their role can be emphasized as:

- They execute the plans of the organization in accordance with the policies and directives of the top management.
- They make plans for the sub-units of the organization.
- They participate in employment & training of lower level management.
- They interpret and explain policies from top level management to lower level.
- They are responsible for coordinating the activities within the division or department.
- It also sends important reports and other important data to top level management.
- They evaluate performance of junior managers.
- They are also responsible for inspiring lower level managers towards better performance.

(c) Lower Level of Management

Lower level is also known as supervisory/operative level of management. It consists of supervisors, foreman, section officers, superintendent etc.

According to R.C. Davis, "Supervisory management refers to those executives whose work has to be largely with personal oversight and direction of operative employees".

In other words, they are concerned with direction and controlling function of management. Their activities include -

- Assigning of jobs and tasks to various workers.
- They guide and instruct workers for day to day activities.
- They are responsible for the quality as well as quantity of production.
- They are also entrusted with the responsibility of maintaining good relation in the organization.
- They communicate workers problems, suggestions, and recommendatory appeals etc to the higher level and higher level goals and objectives to the workers.

Q.5. Explain the social and ethical responsibility of a Manager.

Ans. Social responsibility is defined as the obligation and commitment of managers to take steps for protecting and improving society's welfare along with protecting their own interest. The managers must have social responsibility because of the following reasons:

1. Organizational Resources - An organization has a diverse pool of resources in form of men, money, competencies and functional expertise. When an organization has these resources in hand, it is in better position to work for societal goals.

2. Precautionary measure - If an organization lingers on dealing with the social issues now, it would land up putting out social fires so that no time is left for realizing its goal of producing goods and services. Practically, it is more cost-efficient to deal with the social issues before they turn into disaster consuming a large part if managements time.

3. Moral Obligation - The acceptance of managers' social responsibility has been identified as a morally appropriate position. It is the moral responsibility of the organization to assist solving or removing the social problems.

4. Efficient and Effective Employees - Recruiting employees becomes easier for socially responsible organization. Employees are attracted to contribute for more socially responsible organizations. For instance - Tobacco companies have difficulty recruiting employees with best skills and competencies.

5. Better Organizational Environment - The organization that is most responsive to the betterment of social quality of life will consequently have a better society in which it can perform its business operations. Employee hiring would be easier and employee would of a superior quality. There would be low rate of employee turnover and absenteeism. Because of all the social improvements, there will be low crime rate consequently less money would be spent in form of taxes and for protection of land. Thus, an improved society will create a better business environment.

But, manager's social responsibility is not free from some criticisms, such as -

1. High Social Overhead Cost - The cost on social responsibility is a social cost which will not instantly benefit the organization. The cost of social responsibility can lower the organizational efficiency and effect to compete in the corporate world.

Cost to Society - The costs of social responsibility are transferred on to the society and the society must bear with them.

Lack of Social Skills and Competencies - The managers are best at managing business matters but they may not have required skills for solving social issues.

Profit Maximization - The main objective of many organizations is profit maximization. In such a scenario the managers decisions are controlled by their desire to maximize profits for the organizations shareholders while reasonably following the law and social custom.

Q.6. Explain the relevance of macroeconomic environment in global business.

Ans. A macro environment refers to the overall, broader economy and the forces affecting it versus a microenvironment, which focuses on a specific sector or region's economy. There are macroeconomic conditions or factors that affect how all businesses operate, which, in turn, affect the economy as a whole. In general, macroeconomics deals with:

- Spending
- Price levels
- Aggregate production

Macroeconomic factors affect companies that are more dependent on the overall health of the economy. Industries involving the production and distribution of staple goods and services tend to function more independently. It is cyclical industries – which largely involve luxury goods and travel – that are more heavily affected by macroeconomic factors.

- A macro environment involves forces that affect the larger economy.
- A company should regularly perform a DEPEST analysis in order to best serve customers and the economy as a whole.
- DEPEST refers to the six broad factors affecting the macroeconomy – Demographic, Ecological, Political, Economic, Socio-cultural, and Technological.

Analyzing a Macro Environment

In order to strategically manage a company, analysts often perform a DEPEST analysis, which identifies the demographical, ecological, political, economic, socio-cultural, and technological factors in the macro environment that can affect how companies operate.

It is important to break down what each of the factors entails to get a better idea of the factors that shape how companies operate, and, therefore, shape the macro environment.

(a) Demographic Factors

- Demographic forces ultimately involve human populations that patronize businesses and contribute to the economy. There are a variety of statistics that must be analyzed when it comes to demography, including age, gender, size, occupation, and need.
- Ultimately, fulfilling the needs of any demographic is the reason that businesses operate in the first place and why they are ultimately successful. Demographics, then, ultimately drive how a company chooses to operate and how broader markets develop.

• Population growth and resulting changes to demographics are important areas of study for any business. A company needs to understand exactly who is patronizing it to best be of service, and thereby, generate a profit.

• Marketing, for any business, depends heavily on paying close attention to changes in demographics. Knowing how to target new or emerging demographics is critical. Changes also must be passed on to operations and production to make sure the company continually meets its clients' needs.

(b) Ecological Factors

• Ecological factors are natural forces that shape a macro environment. It is largely shaped by the ease of access to natural resources used in the production and distribution of a company's goods and services.

• Environmental concerns affect both access to natural resources and the supply of natural resources a company can utilize. As populations expand, access to natural resources increases, which tends to lead to a depletion of said resources. Pollution increases resulting from population growth also shape the environment.

• For years, there's been a growing shortage of raw materials, and a heavier emphasis is being placed on changing business practices to utilize more sustainable materials. The hope is that placing less burden on the environment will help to regenerate ecological cycles and make materials both more plentiful and more accessible.

(c) Political Factors

• Businesses are always limited by the political environment in which they exist. Laws and governments regularly shape how a company can operate and even have sway over the markets that companies can serve.

• The primary instance where political factors play a critical influence is when a company tries to move into a new market – specifically one in a different country. The company must understand the laws and regulations that dictate both the industry it operates in and any specific rules it may be subjected to. It may affect whether it's financially responsible for the company to enter the new market at all.

• It's also important to keep abreast of new legislation in the works or tabled for passage. Understanding shifting political forces can help a company change directions to skirt any serious impact should new regulations or rules be passed.

(d) Economic Factors

• Economic factors affecting the macro environment relate to forces that affect how consumers spend and their purchasing power. It is important to understand a variety of metrics and data, including:

- Gross Domestic Product (GDP) and its real growth rate
- Unemployment rates
- Inflation
- Disposable personal income
- Existing spending patterns

• Every business should closely monitor data regularly and be fully aware of such numbers before moving into new markets in different countries.

(e) Socio-Cultural Factors

Socio-cultural factors relate to demographics in a sense but are more related to populations and how they behave based on preference and values. Different societies and cultural groups are characterized by different needs, which are often based on different core values and preferences.

Cultures often develop a group mentality, which passes along core values and general beliefs. It shapes how the individuals in such cultures shop and what they choose to spend their money on. A business needs to pay attention to socio-cultural variances, especially when moving into new markets.

Technological Factors

Technological factors refer to the creation of new technologies and how they shape products, product development, and access to new market opportunities. A perfect example of a strong technological force today is wireless communication.

Nearly everyone in the world owns a smartphone, tablet, or laptop that can quickly and easily be connected to the internet. It exerts a serious impact on individuals and societies and their ability to patronize companies freely.

Mobile technology is also shaping the development of new technological devices and replacing ones that have become outdated. A company needs to continually use the most up-to-date technology in order to operate at its highest capacity and be aware of how technological applications can better serve customers.